

A

Project Report

on

## “Sign Language to Text and Speech”

Submitted in partial fulfillment of requirements for the award of Degree of

Bachelor of Technology in

[Computer Technology]

**SECOND YEAR**

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**Established by Maharashtra Act VII of 2021 dated 23rd March 2021**

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**A. Y. 2024-25**



## **RATAN TATA MAHARASHTRA STATE SKILLS UNIVERSITY**

**Established by Maharashtra Act VII of 2021 dated 23rd March 2021**

### **DEPARTMENT OF COMPUTER TECHNOLOGY**

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## **STUDENT'S DECLARATION**

We hereby declare that the work being presented in this project entitled “Sign Language to Text and Speech” in partial fulfilment for the award of degree of Bachelor of Technology in Computer Technology and submitted at the Department of Computer Technology of Ratan Tata Maharashtra State Skills University is an authentic record of our work carried out under the guidance of Prof. Harpreet Khurana and Prof. Manish Agrawal, HoD, Department of Computer Technology, Ratan Tata Maharashtra State Skills University, Mumbai.

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This is to certify that the above declaration made by the candidate is correct and best of our knowledge. Examination of this work is carried on \_\_\_\_\_

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## **LIST OF ACRONYM**

| <b>Acronym</b> | <b>Full Form</b>             |
|----------------|------------------------------|
| <b>DHH</b>     | Deaf and Hard-of-Hearing     |
| <b>ASL</b>     | American Sign Language       |
| <b>CNN</b>     | Convolutional Neural Network |
| <b>TTS</b>     | Text To Speech               |
| <b>DFD</b>     | Data Flow Diagram            |
| <b>ROI</b>     | Region Of Interest           |
| <b>GUI</b>     | Graphical User Interface     |

## Acknowledgement

We would like to express our sincere gratitude to all those who have contributed to the successful completion of this project entitled “Sign Language to Text and Speech”, in partial fulfilment for the award of degree of Bachelor of Technology in Computer Technology.

We are deeply indebted to our project guides, Prof. Harpreet Khurana and Prof. Manish Agrawal, HoD, Department of Computer Technology, Ratan Tata Maharashtra State Skills University, Mumbai, for their invaluable guidance, constant encouragement, insightful suggestions, and unwavering support throughout the duration of this project. Their expertise and motivation were instrumental in shaping this work.

We also extend our thanks to the Department of Computer Technology, Ratan Tata Maharashtra State Skills University for providing the necessary resources, infrastructure, and conducive environment for carrying out this research.

This project was significantly inspired and guided by the foundational work in the field. We particularly acknowledge the base paper by Bikash Yadav and Hasan Bohra, alongside other relevant studies, which provided crucial insights into the challenges and requirements of robust sign language recognition systems, particularly emphasizing the importance of data diversity and real-time performance.

Without their collective support, this project would not have been possible.



## Abstract

This project addresses the pressing communication challenges that Deaf and Hard-of-Hearing (DHH) communities encounter daily when interacting with individuals unfamiliar with sign language. Such barriers can lead to limited access to education, healthcare, and employment opportunities, highlighting the need for a practical, real-time solution. The primary objective here is to design and implement a user-friendly system capable of detecting and translating American Sign Language (ASL) gestures into textual and spoken outputs, thereby enabling seamless, two-way interaction. By emphasizing universal accessibility, the initiative aspires to reduce communication gaps and foster greater inclusion across public, professional, and educational spheres.

At the core of the methodology is a computer vision pipeline that uses a standard camera to continuously capture frames, isolating the hand region and its distinctive geometry. Data preprocessing includes background removal, contour detection, and skin-color segmentation, ensuring that essential hand features are distinctly highlighted. These refined frames are fed into a convolutional neural network (CNN) trained on a curated dataset of ASL gestures. The network learns representative features through multiple convolution and pooling layers, culminating in a fully connected layer for classification. As each gesture is interpreted, the system composes letters into complete words or sentences, which are then converted into spoken language via text-to-speech (TTS) technology.

Initial trials indicate that the model effectively identifies static ASL gestures under moderate variations in lighting and hand positioning. Results suggest that larger and more heterogeneous datasets—encompassing different skin tones, complex backgrounds, and faster signing speeds—would further strengthen the system’s accuracy and resilience. Beyond classification precision, user-centric considerations such as responsiveness and intuitive interface design are pivotal to ensuring a smooth experience for both signers and listeners. Future advancements may include extending recognition to dynamic gestures, exploring context-based language models for more fluent sentence creation, and integrating additional modules for automated punctuation or error correction.

In summary, this project integrates deep learning and computer vision to translate ASL gestures into text and speech, bridging a substantial communication gap for DHH individuals. By overcoming key limitations related to limited vocabulary, diverse environmental conditions, and real-time performance, the proposed approach holds promise for improving accessibility and inclusivity—enabling Deaf and Hard-of-Hearing users to effectively communicate with non-signers across various social, educational, and professional settings.

# Chapter 1

## 1. Introduction

Communication barriers present a significant challenge for Deaf and Hard-of-Hearing (DHH) individuals, often restricting their ability to interact seamlessly in public, professional, and educational contexts. While sign language is a primary mode of communication within DHH communities, it remains largely unfamiliar to the majority of the hearing population. Consequently, the need for interpreters or alternative communication methods can become a daily hurdle, hindering personal autonomy and limiting opportunities in various sectors. These constraints reveal a profound demand for technological solutions that can bridge the linguistic gap and enable inclusive communication channels.



### 1.1 Problem and Motivation

The root of this problem lies in the substantial shortage of ASL-fluent interpreters available at all times and places. Although professional interpreters can provide reliable solutions, they are not always accessible or affordable, especially for hasty or brief interactions—such as clarifying instructions at a hospital counter, seeking help in a retail store, or conversing in social gatherings. Even in venues where interpreters are offered, consistent scheduling or service fees might deter meaningful participation from DHH individuals. Handwritten notes or text apps, while somewhat helpful, tend to lose the expressive nuance that instant communication provides.

Additionally, most mainstream natural language tools have traditionally concentrated on spoken or typed languages, leaving sign languages less explored. By designing a sign-language-to-text-and-speech converter, our project responds to an underrepresented need, opening the door to greater autonomy for DHH users. Crucially, the system utilizes

commonly available hardware—such as standard webcams or smartphone cameras—making it feasible to deploy in a wide array of real-world locations without major adjustments.

## **1.2 Technical Foundations**

In general terms, the system leverages commonly adopted computer vision techniques to detect and isolate the signer's hands from a video feed. Methods such as background subtraction or color-based segmentation help remove noise, ensuring the hand gesture is clearly visible to the classifier. These segmented frames are then directed into a convolutional neural network (CNN) primed to recognize specific ASL signs. CNNs excel at parsing visual patterns and identifying distinctions in shape or angle that characterize certain gestures.

Once a sign is classified, the system maps the recognized gesture to a corresponding letter or word. The letters may then be combined into words, which are displayed on-screen and optionally vocalized using text-to-speech (TTS) libraries. This loop permits a signer to communicate with hearing individuals, who hear the message in spoken form. Low latency and robust classification accuracy are essential to delivering seamless conversations—requirements that necessitate consistent testing with diverse datasets, lighting conditions, and user demographics.

## **1.3 Implementation Snapshot**

When a user positions themselves in front of the camera, each video frame is captured, processed, and classified in near real time. As soon as a letter or word is identified, it is appended to a live text display. Simultaneously, text-to-speech functionality can announce the recognized words, bridging the communication gap between signers and those who do not know sign language.

Various challenges emerge in practice: changes in lighting, multiple skin tones, and background complexity can mislead the model. To address these, a large and varied training dataset is necessary—one that includes different hand shapes, skin colors, and environmental conditions. Future expansions may incorporate dynamic ASL gestures, along with facial expression detection, which can add further context to many signed statements.

## **1.4 Relevance and Future Prospects**

Deploying a robust sign language translator can transform multiple fields—education, healthcare, customer service—by making spontaneous interactions more accessible. For

instance, in a school environment, an “on-demand” translator can improve inclusivity with minimal downtime. Healthcare providers, too, benefit from immediate sign-to-speech conversions, allowing for direct patient-provider communication without the logistical complexity of scheduling interpreters.

As computer vision and deep learning frameworks evolve, sign language recognition will likely benefit from faster inference, more extensive datasets, and advanced network architectures. Further goals include supporting multiple sign languages and incorporating motion tracking to capture the nuances of dynamic gesture transitions. Altogether, this project demonstrates that modest equipment (e.g., a typical webcam) and targeted software can significantly enhance communication for DHH individuals, reducing the everyday barriers that limit their full participation in society.

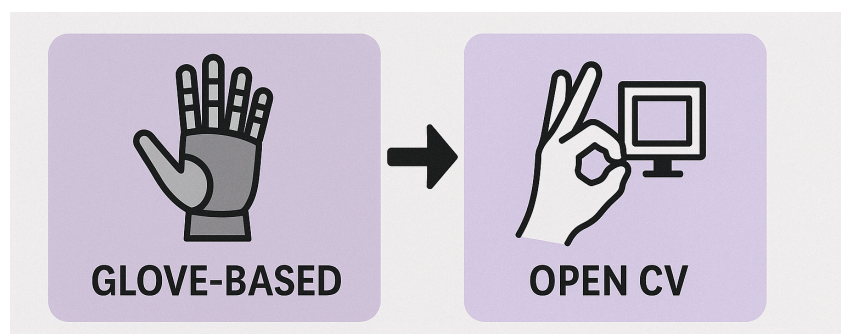
## Chapter 2

### 2. LITERATURE REVIEW

Communication barriers frequently prevent Deaf and Hard-of-Hearing (DHH) individuals from fully engaging in public, professional, and educational contexts. Although sign languages remain the primary medium of communication within these communities, most hearing individuals lack proficiency in them. In response, numerous studies have investigated various ways—ranging from sensor-based gloves to purely computer vision systems—to automate sign language recognition. The “base paper” (attached as “base paper.pdf”) presents a detailed overview of vision-based hand gesture classification using convolutional neural networks (CNNs), emphasizing the importance of accurate feature extraction and robust background segmentation.

#### 2.1 Related Works and Their Relevance

Early sign language recognition approaches often depended on specialized gloves equipped with sensors to record hand motion and finger flexion, which, although accurate, were impractical for everyday use due to high costs and cumbersome setups. Over time, research shifted toward camera-based detection that utilized classical image processing methods like thresholding, background subtraction, and contour detection. This evolution reflected the desire for nonintrusive systems that can detect hand movements or shapes without the need for additional hardware. The base paper aligns with this trend by highlighting segmentation and feature extraction techniques crucial for achieving high recognition rates in real-world conditions.



Subsequent contributions introduced deep learning frameworks, particularly CNNs, to significantly boost classification accuracy. By progressively learning from raw image data, CNNs can identify complex and subtle differences among signs. Transfer learning methods further streamlined training, demonstrating how pre-trained models could be adapted to sign

language tasks with relatively small datasets. Integrating text-to-speech (TTS) mechanisms later emerged as a natural extension of these studies, allowing direct auditory output that helps hearing individuals receive messages without reading off a screen.

## **2.2 Project Requirements**

Guided by the findings in the base paper by Bikash Yadav and Hasan Bohra and other relevant works, a robust sign language recognition system must offer dependable hand detection, precise classification capabilities, and minimal latency for real-time interaction. A crucial insight drawn from these studies is the value of training data diversity, whether generated through data augmentation (rotations, flips, or shifts) or collected in varied lighting conditions and backgrounds. Adapting the model to handle multiple hand shapes, skin tones, and signing styles remains an ongoing challenge, yet it is essential for broadening the applicability of the technology. Equally important is ensuring that recognized letters, words, or phrases can be converted into more user-friendly outputs, such as text displayed on-screen or synthesized speech.

## **2.3 Methodological Insights and Gaps**

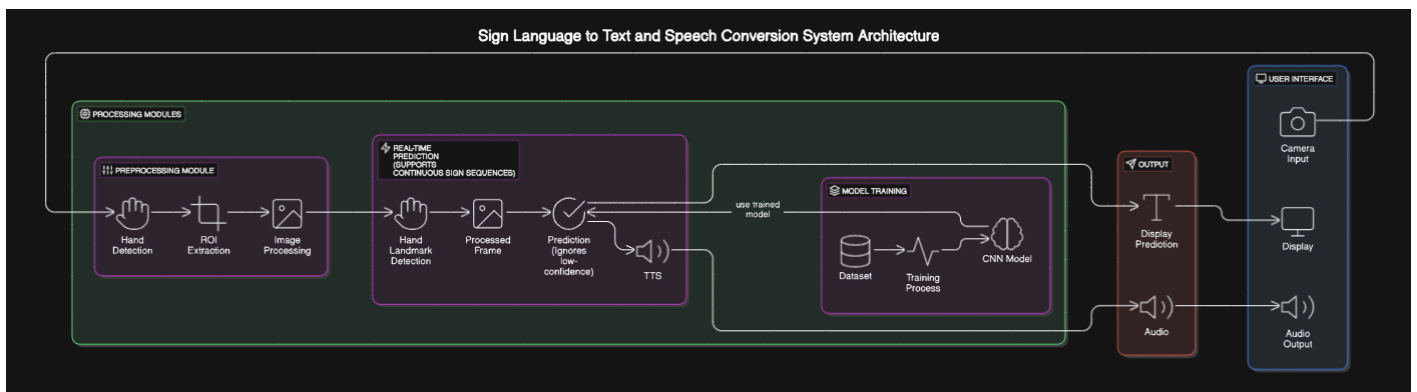
Methodologically, the literature underscores that hand region isolation significantly influences model performance. Complex or changing backgrounds, as well as variations in lighting, can degrade accuracy if preprocessing is inadequate. Neural network models benefit from clear foreground segmentation, often achieved through thresholding or color-based filtering, and from additional enhancements like skeleton-based keypoint estimation (e.g., using frameworks such as Mediapipe). Despite these advancements, researchers note inherent gaps when moving beyond static alphabets to dynamic or phrase-level signing. The base paper discusses how future directions could involve extended vocabularies and dynamic gesture recognition capable of handling transitions between signs. Our project aims to address these concerns through a combination of CNN classification, hand landmarking, and real-time text-to-speech output, thereby expanding the system's usability across a variety of environments without incurring high computational costs or requiring specialized hardware.

## Chapter 3

# Design of System/Model

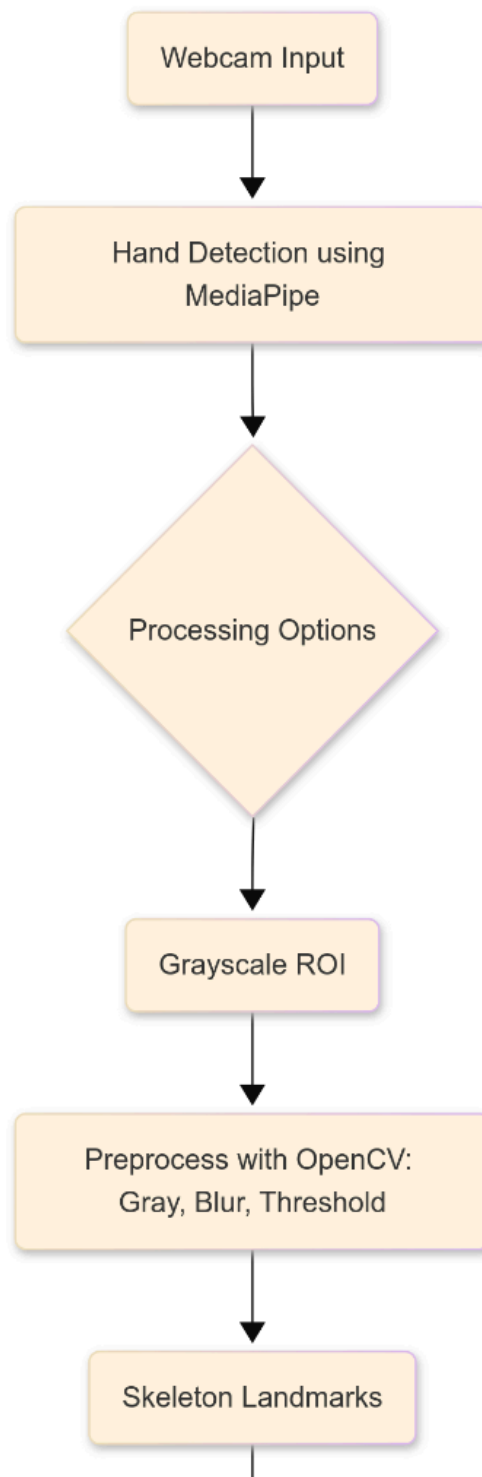
### 3.1 System Architecture

A system architecture diagram provides a high-level view of how the core components interact. Typically, it shows layers like the user interface (front-end), business logic (recognition or processing module), and data storage (database or cloud service). By outlining these parts, viewers can quickly grasp where data originates (e.g., webcam input), how it's processed (e.g., gesture recognition engine), and where it's stored or delivered (e.g., text output on a screen or speech synthesis).

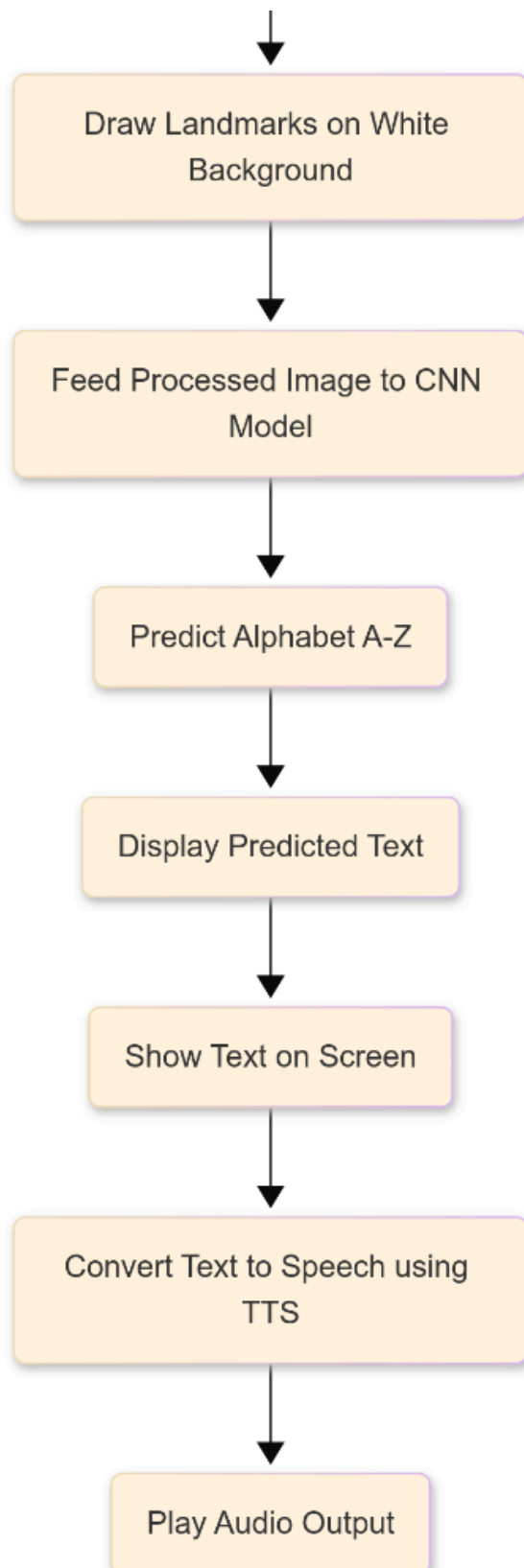


## 3.2 System Flowchart

A flowchart clarifies the logical steps of your system from start to finish. It often begins with user input—for instance, a sign language gesture—and follows decision points such as “Is hand detected?” or “Is sign recognized?” before concluding in outcomes like “Display text” or “Provide speech output.” Each step is represented by shapes (rectangles for processes, diamonds for decisions), showing how the system transitions or branches in different scenarios.

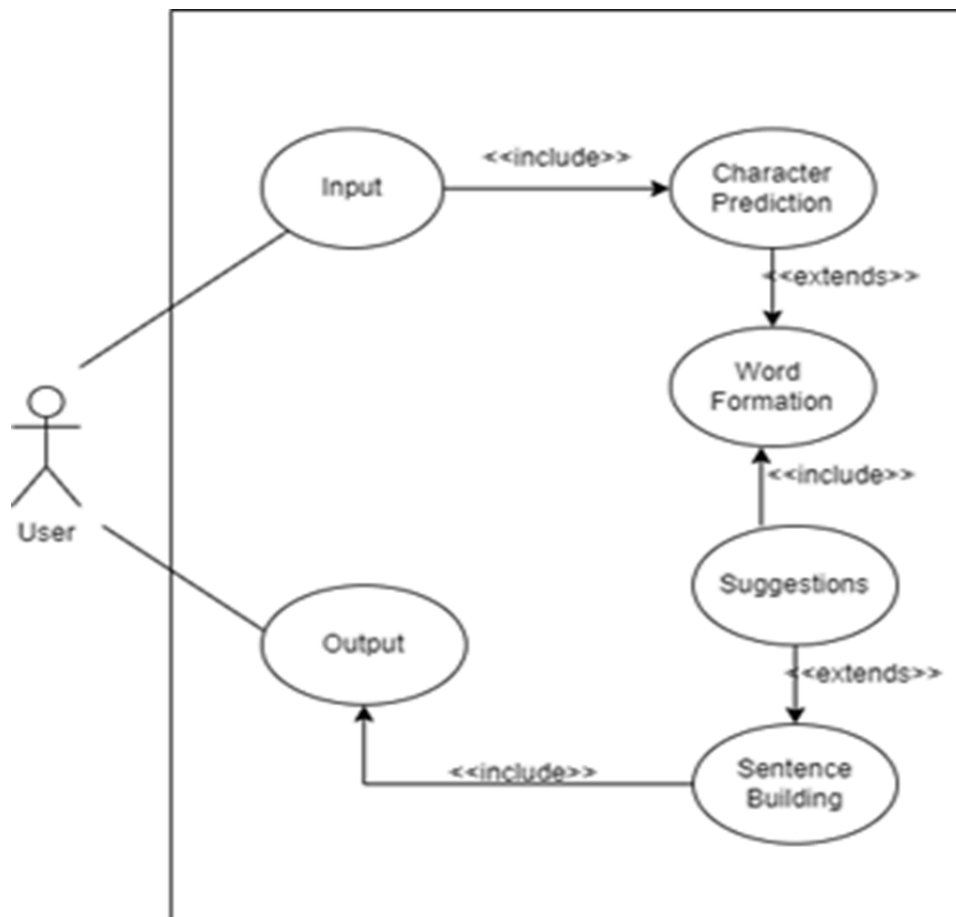






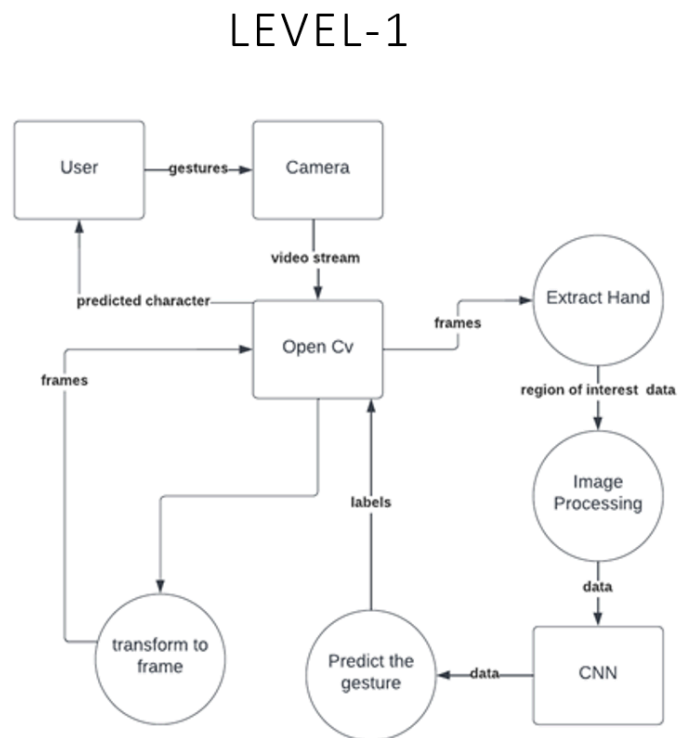
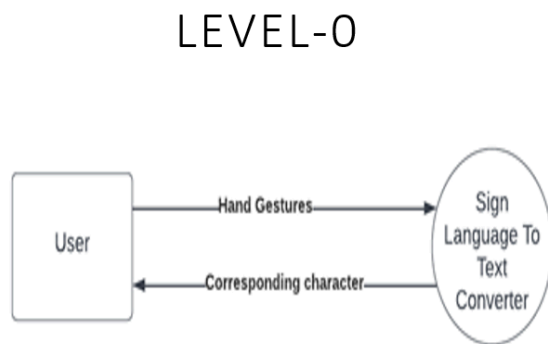
### 3.3 Use Case Diagram

A use case diagram depicts actors (like “User” or “System Admin”) and their interactions with the system’s functionalities. For sign-language-to-text conversion, a typical use case might be “Recognize Gesture,” which the User initiates. Lines connecting an actor to a particular use case indicate that the actor can trigger or benefit from that functionality. This diagram ensures clarity about who uses the system and in what capacity, highlighting the primary actions involved (such as “Initiate Detection” or “Receive Speech Output”).



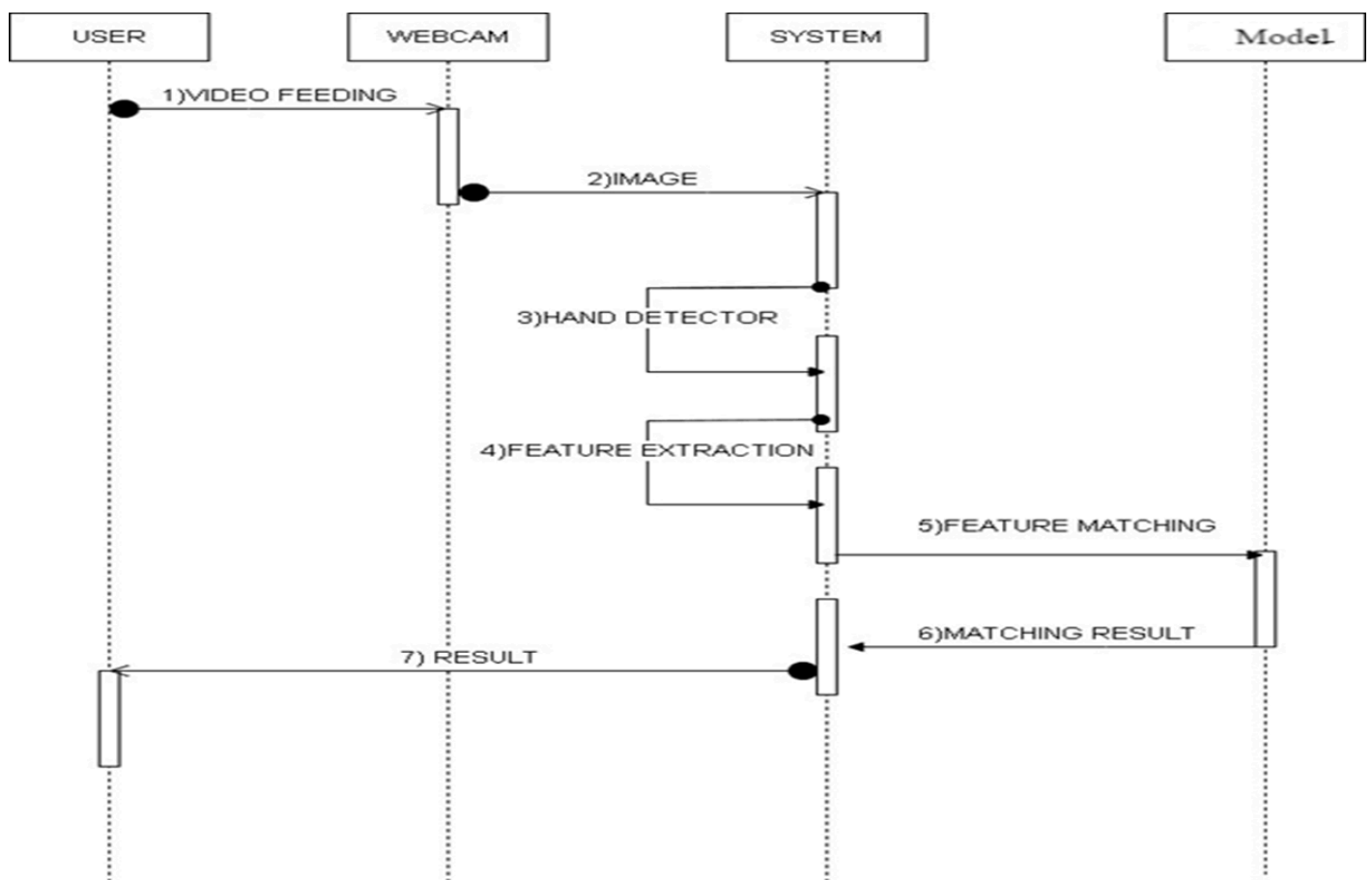
### 3.4 Data Flow Diagram (DFD)

A DFD captures how data traverses the system and transforms at each stage. In a Level 0 (context) diagram, you typically represent the entire system as a single process with arrows pointing to or from external entities (e.g., “User,” “Database”). At deeper levels (Level 1, Level 2), you break that process into smaller sub-processes—like “Capture Hand Gesture,” “Classify Sign,” “Store/Retrieve Data,” and “Generate Speech.” This helps illustrate exactly where and how data is moving, stored, and acted upon within the architecture.



### 3.5 Sequence Diagram

A sequence diagram shows the order of operations among system objects or components over time. Vertical lines represent different components (e.g., User, Recognition Module, TTS Engine), while horizontal arrows convey the messages or function calls exchanged in a specific scenario—such as “User initiates sign recognition,” “System classifies gesture,” and “TTS provides audio output.” This format helps pinpoint which processes occur sequentially and how data or control moves from one component to another, clarifying the real-time interaction across modules.



## **Chapter 4**

### **Working of System/Model**

#### **4.1 Data Acquisition**

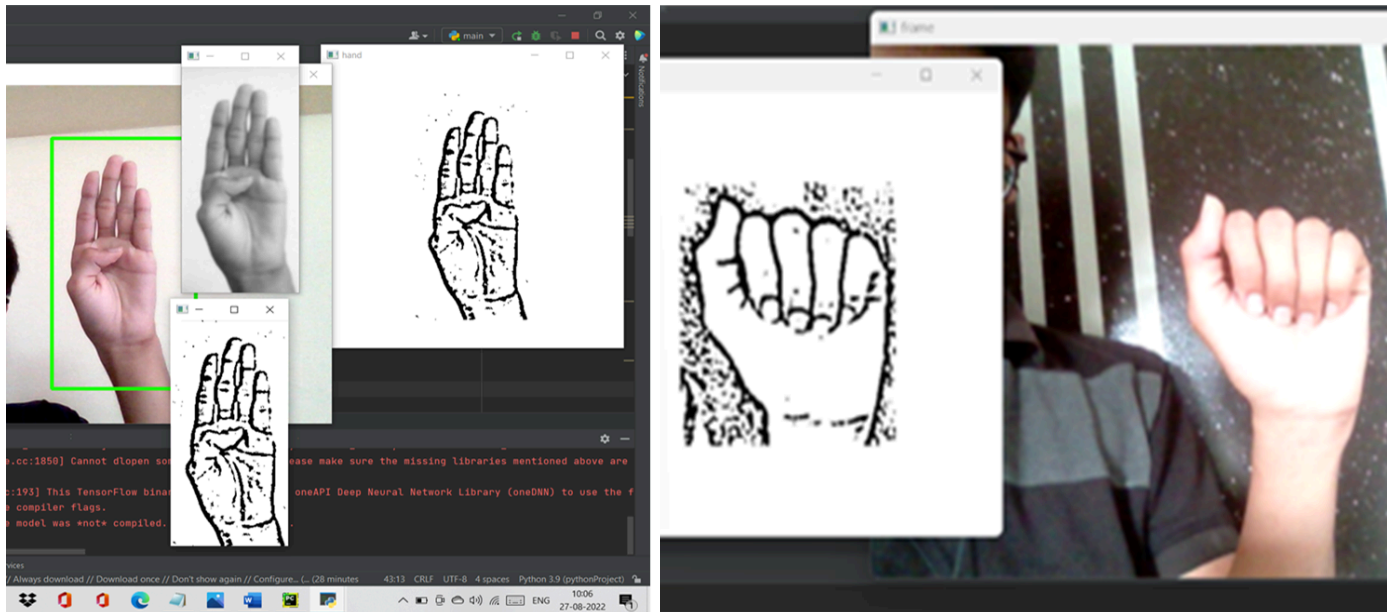
The different approaches to acquire data about the hand gesture can be done in the following ways:

It uses electromechanical devices to provide exact hand configuration and position. Different glove-based approaches can be used to extract information, but it is expensive and not user-friendly.

In vision-based methods, the computer webcam is the input device for observing the information of hands and/or fingers. The vision-based methods require only a camera, thus realizing a natural interaction between humans and computers without the use of any extra devices, thereby reducing costs. The main challenge of vision-based hand detection ranges from coping with the large variability of the human hand's appearance due to a huge number of hand movements, to different skin-color possibilities, as well as to the variations in viewpoints, scales, and speed of the camera capturing the scene.

#### **4.2 Data Preprocessing and Feature Extraction**

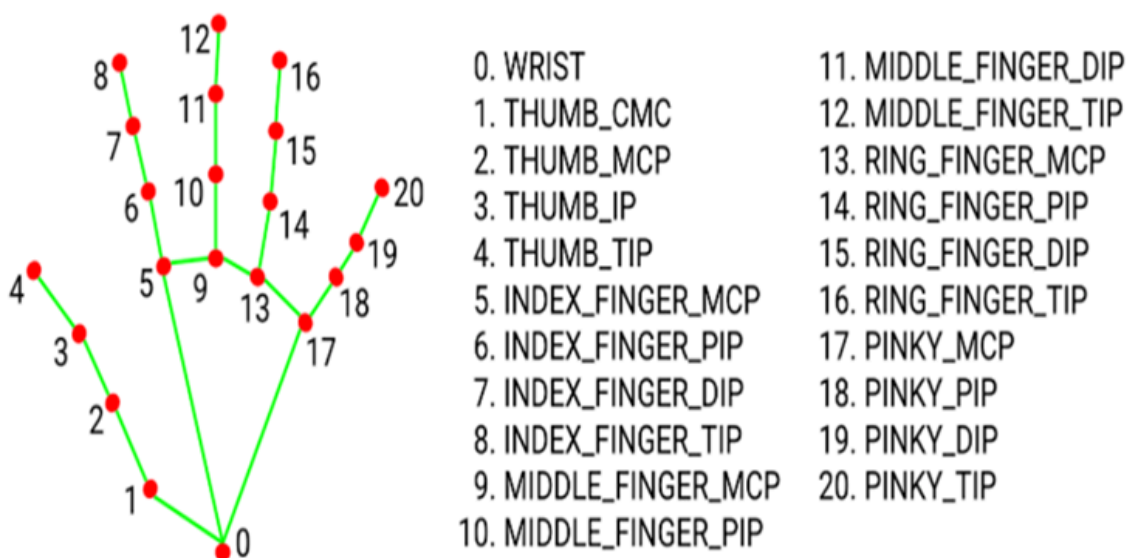
In this approach for hand detection, firstly we detect the hand from the image that is acquired by the webcam. For detecting a hand, we used the MediaPipe library, which is used for image processing. So, after finding the hand from the image, we get the Region of Interest (ROI). Then we cropped that image and converted it to a gray image using the OpenCV library. After we applied the Gaussian blur, the filter can be easily applied using the Open Computer Vision library, also known as OpenCV. Then we converted the gray image to a binary image using threshold and adaptive threshold methods. We have collected images of different signs from different angles for sign letters A to Z.

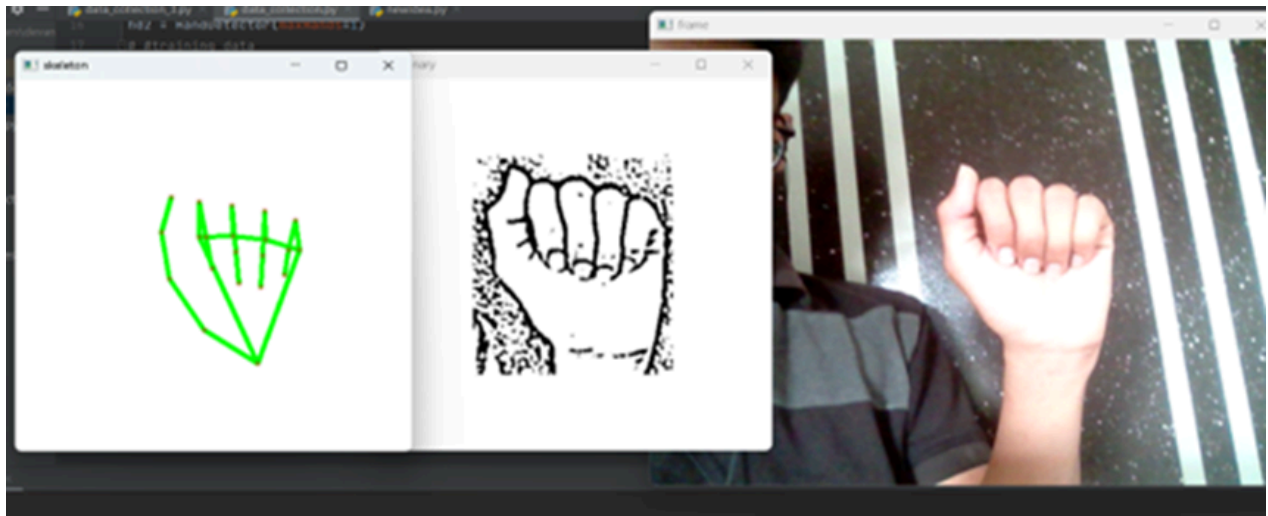
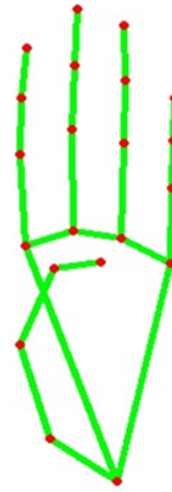


In this method, there are many loopholes. For example, your hand must be ahead of a clean soft background and in proper lightning conditions. Only then will this method give good accurate results. But in the real world, we don't get good backgrounds everywhere, and we don't get good lightning conditions either.

So, to overcome this situation, we tried different approaches. Then we reached one interesting solution in which, firstly, we detect the hand from the frame using MediaPipe and get the hand landmarks present in that image. Then we draw and connect those landmarks in a simple white image.

## Mediapipe Landmark System





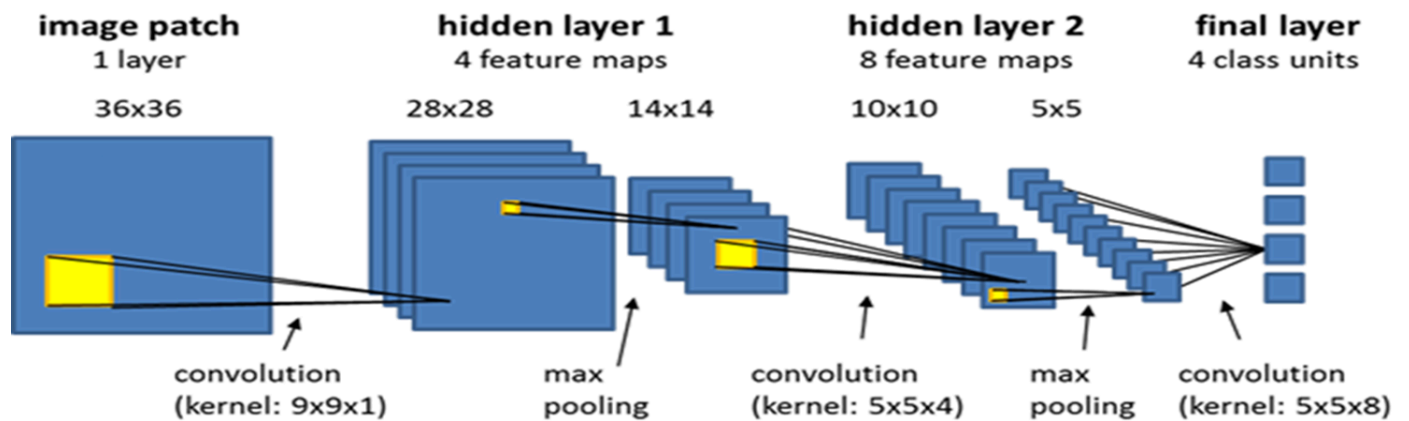
Now we will get these landmark points and draw them in a plain white background using the OpenCV library. By doing this, we tackle the situation of background and lightning conditions because the MediaPipe library will give us landmark points in any background and mostly in any lightning conditions. We have collected 180 skeleton images of alphabets from A to Z.

## 4.3 Gesture Classification

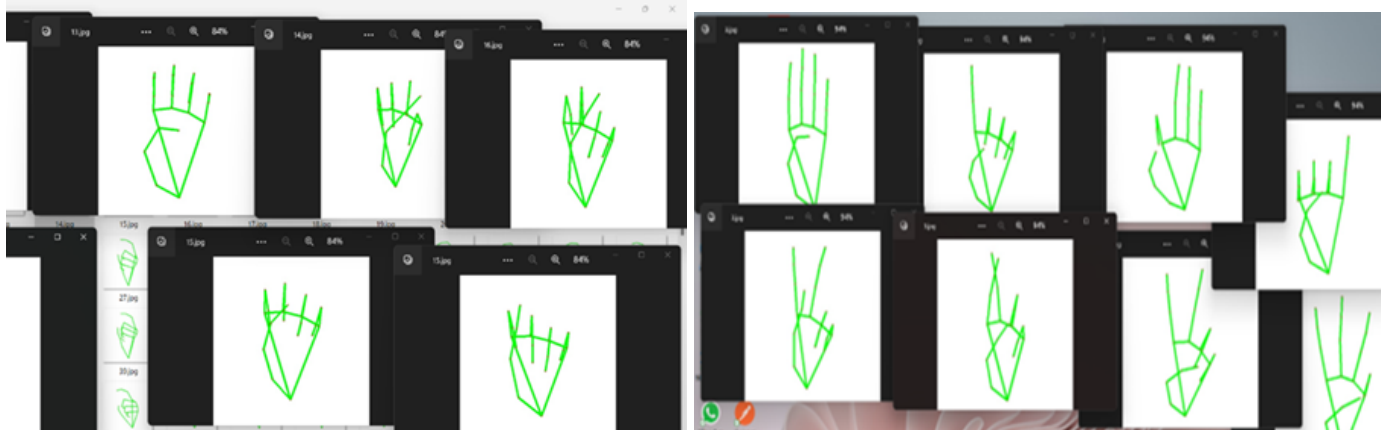
### Convolutional Neural Network (CNN)

CNN is a class of neural networks that are highly useful in solving computer vision problems. They found inspiration from the actual perception of vision that takes place in the visual cortex of our brain. They make use of a filter/kernel to scan through the entire pixel values of the image and make computations by setting appropriate weights to enable detection of a specific feature. CNN is equipped with layers like convolution layer, max pooling layer, flatten layer, dense layer, dropout layer, and a fully connected neural network layer. These

layers together make a very powerful tool that can identify features in an image. The starting layers detect low-level features that gradually begin to detect more complex higher-level features. By the end of the CNN architecture, we will reduce the full image into a single vector of class scores.



The preprocessed 180 images/alphabet will feed the Keras CNN model. Because we got bad accuracy in 26 different classes, thus, we divided the whole 26 different alphabets into 8 classes, in which every class contains similar alphabets:



All the gesture labels will be assigned with a probability. The label with the highest probability will be treated as the predicted label. So, when the model classifies [AEMNST] in one single class using mathematical operation on hand landmarks, we will classify further into the single alphabet A, E, M, N, O, R, S, or T.



## 4.4 Text and Speech Translation

The model translates known gestures into words. We have used the `pyttsx3` library to convert the recognized words into the appropriate speech. The text-to-speech output is a simple workaround, but it's a useful feature because it simulates a real-life dialogue.

## Chapter 5

### Result & Discussion

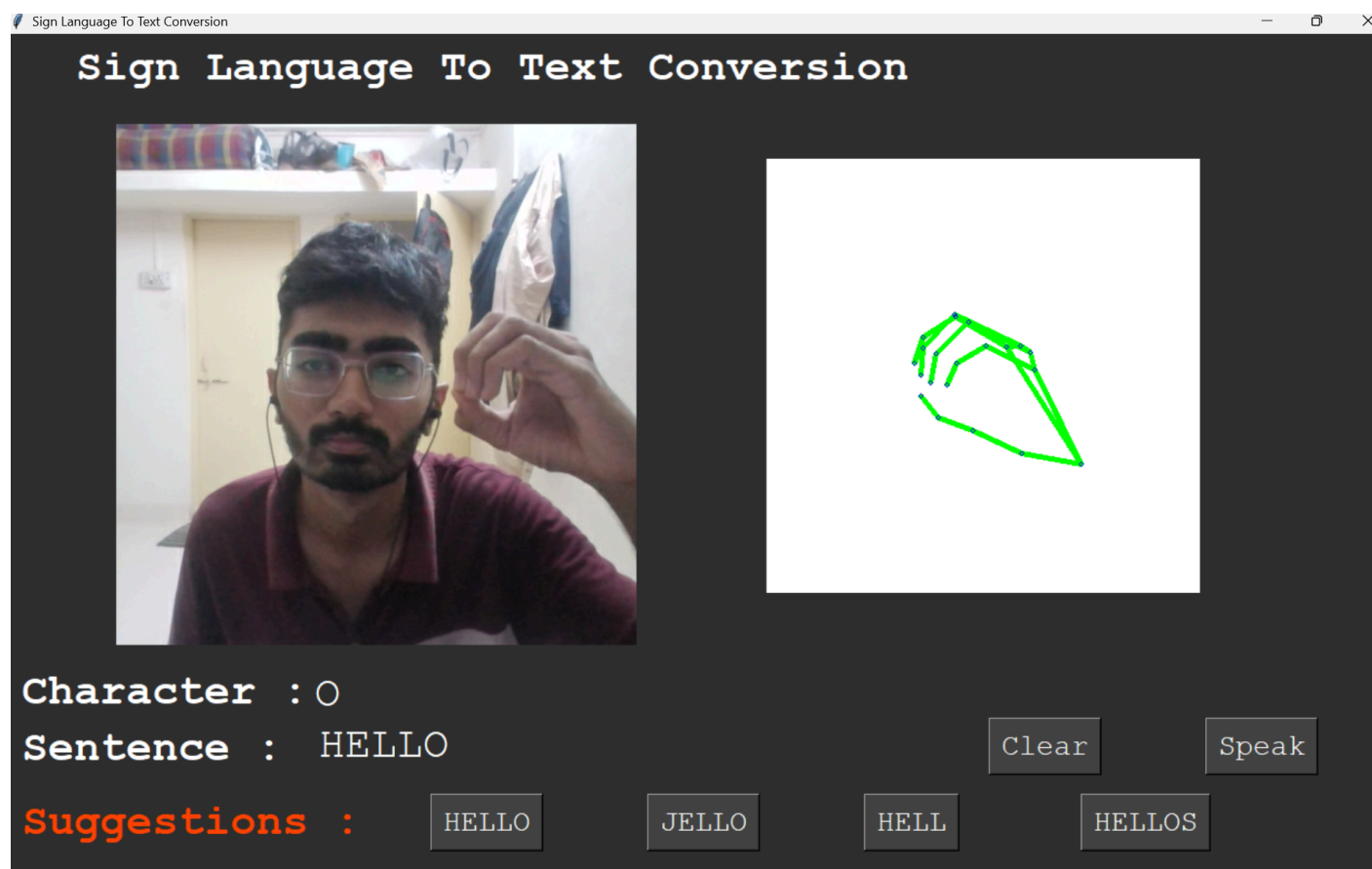
The implemented system successfully translates American Sign Language hand gestures into text and speech. The core implementation involves capturing hand gestures via a webcam feed and processing these images to identify the corresponding alphabet.

A crucial aspect of the system's success lies in the data pre-processing and feature extraction phase. Initially, traditional methods relying on background subtraction or contour analysis were explored but proved sensitive to variations in background and lighting conditions, yielding inconsistent results in real-world scenarios. To overcome this, the project adopted a more robust approach utilizing the MediaPipe library. This library accurately detects hand landmarks regardless of the background or typical lighting variations, generating skeleton images representing the hand's pose. By training the model on these landmark-based skeleton images rather than raw hand images or contours, the system effectively focuses on the gesture structure itself, making it significantly more resilient to environmental factors. We collected and preprocessed 180 skeleton images per alphabet (A-Z) to build a comprehensive dataset.

For gesture classification, a Convolutional Neural Network (CNN) model built with Keras was employed. The model was trained on the preprocessed skeleton images. While a direct classification into 26 distinct classes (A-Z) faced challenges, leading to lower initial accuracy, refining the approach by grouping similar-looking gestures into 8 broader classes significantly improved performance. Subsequent classification within these broader classes, potentially using hand landmark relationships, refined the prediction to a specific alphabet.

Testing demonstrated the system's ability to correctly predict hand gestures for the American Sign Language alphabet in various backgrounds and lighting conditions, as evidenced by the implementation snapshots. A user-friendly Graphical User Interface (GUI) was developed using Python's Tkinter library to facilitate interaction. This interface displays the webcam feed, the processed skeleton image, and the predicted character/sentence. Usability features were included, such as the incorporation of specific hand signs to represent actions like adding a space between words or confirming a predicted character to build a sentence. A

suggestion feature was also added to provide alternative predictions, smoothing the user experience.



The overall results were highly promising. The system achieved an impressive accuracy of 97% in predicting alphabets (A-Z) across various testing conditions, including those with complex backgrounds and imperfect lighting. Furthermore, in optimal conditions with clean backgrounds and good lighting, the accuracy reached an even higher 99%. This high level of accuracy directly validates the effectiveness of the chosen methodology, particularly the use of MediaPipe hand landmark detection and the subsequent CNN classification on the invariant skeleton representations, which effectively tackles the variability inherent in real-world hand gesture recognition. The combination of a robust feature extraction method and a well-trained classification model resulted in a system capable of reliable sign language interpretation into text and speech.

## Chapter 6

### CONCLUSION

This project successfully developed and implemented a robust vision-based system for translating American Sign Language (ASL) hand gestures into text and speech. The system's primary objective is to serve as a vital communication bridge, holding immense potential to significantly benefit the deaf and dumb community. By enabling easier and more fluid interaction with individuals who do not understand sign language, this technology directly addresses critical communication barriers. This facilitates everyday conversations, opens doors to greater educational and employment opportunities, and improves access to essential services, ultimately fostering enhanced social inclusion and independence for millions worldwide.

A key challenge in vision-based sign language recognition is reliably handling real-world environmental variability, particularly diverse backgrounds and fluctuating lighting conditions. We effectively addressed this by strategically integrating the powerful MediaPipe library, known for its accurate hand landmark detection regardless of the surroundings. By extracting these specific landmark points and using them to generate standardized skeleton images, the system focuses on the essential structure of the hand gesture itself, creating robust features invariant to external conditions. This technical choice was crucial for ensuring the system's reliability and practicality in real-world usage.

The core of our gesture recognition engine is a Convolutional Neural Network (CNN) model, trained on a comprehensive dataset derived from MediaPipe skeleton images. We found that refining the classification strategy from attempting 26 distinct alphabet classes to initially grouping similar-looking gestures into 8 broader categories significantly improved the model's performance and efficiency. Subsequent, more refined classification within these broader groups allowed for precise identification. Testing demonstrated the system's reliability and accuracy, achieving an impressive 97% overall under diverse conditions and an outstanding 99% in optimal settings. These high figures are critical for a dependable communication tool, instilling user confidence.

Finally, a user-friendly Graphical User Interface (GUI) built with Python's Tkinter enhances the system's accessibility and practicality, offering real-time feedback, intuitive sentence building controls, and helpful prediction suggestions. In conclusion, this project has delivered an accurate and robust sign language translation system that effectively overcomes common technical hurdles and holds significant potential to empower the deaf and dumb community by breaking down communication barriers and promoting greater integration into society.

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- [https://www.ripublication.com/ijaer18/ijaerv13n9\\_90.pdf](https://www.ripublication.com/ijaer18/ijaerv13n9_90.pdf)